The alpha-function tempotron: the biological causal kernel and the powerless-tail correction

Tempotron learning-surface program

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Abstract

The biologically faithful post-synaptic potential (PSP) is the causal alpha-function $K(s) = (s/\tau)\, e^{1-s/\tau}\, \Theta(s)$ (Rall 1967): one-sided, peaking at $s = \tau$. We treat the alpha-kernel tempotron as a reduction in its own right and ask what its biological causality does to capacity. Three facts organise the analysis. (i) The spectrum $\hat{K}(\omega) = e\tau/(1 + i\omega\tau)^2$ is a *second-order* low-pass (Matérn $\nu = 3/2$, $S(\omega) \sim \omega^{-4}$): the spectral moments give a clean effective dimension $K_{\text{eff}} = T/\tau$, but the fourth moment λ_4 *diverges*, so the maxima-count axis is forbidden and only the λ_2-based K_{eff} is admissible. (ii) Because the kernel is one-sided, the membrane process is non-stationary in a boundary layer of width $\sim 3\tau$ at the window edge—the stationary-Gaussian-process approximation is weaker here than for any symmetric kernel, an honesty caveat we carry throughout. (iii) Most sharply, the ω^{-4} tail inflates the Rice K_{eff} (through λ_2) with frequencies that carry negligible *power*: $\sim$82% of the power sits below $\omega = 1/\tau$ but only $\sim$18% of λ_2 does. We test this directly. In a lab capacity sweep the alpha kernel gives findable $\hat{\alpha}_c = 2.00, 2.34, 2.72$ at $K_{\text{eff}} = 16.2, 32.7, 65.7$; stripping the high-frequency tail at $8/\tau$ drops Rice K_{eff} by $\sim$10% (32.7→29.3, 65.7→58.8) but leaves $\hat{\alpha}_c$ unchanged (2.34→2.40, 2.72→2.77, within α-grid resolution). **Capacity tracks low-frequency power, not the tail-inflated** K_{eff}—so the robust expressiveness axis is the participation dimension $D_{\text{eff}} = (\int S)^2 / \int S^2$, with the SymPy-exact ratio $D_{\text{eff}}(\text{alpha})/D_{\text{eff}}(\text{gauss}) = 2\sqrt{\pi}/5 \approx 0.71$ at matched K_{eff}. Honestly, the alpha kernel also sits ~ 0.7 *below* the gaussian baseline at matched K_{eff} (2.34 vs. 3.04 at $K_{\text{eff}} \approx 32$): a causal suppression *stronger* than the slope-preserving "hair" the theory predicted. Status: **PARTIAL**—the I4 truncation result is clean and twice-confirmed; the causal/edge GP approximation is the weakest in the series, the absolute numbers are RS-level, and the over-large causal deficit is unexplained.

Contents

Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Model and setup

Tempotron. A tempotron [5] fires when its peak membrane potential crosses threshold,

$$V_{\max}(\mathbf{w}) \;=\; \max_{t\in[0,T]} \sum_{i=1}^{N} w_i \sum_f K\Big(t - t_i^f\Big) \;>\; \vartheta, \tag{1}$$

where K is the PSP kernel and $\{t_i^f\}$ the presynaptic spikes. The capacity $\alpha_c = P/N$ is the load at which a fraction $\frac{1}{2}$ of random balanced ± 1-labelled patterns can be separated; the linear (single-time-sample) anchor is the Cover/Gardner perceptron value $\alpha_c = 2$ [2, 3].

The biological causal kernel. Every other kernel in this program (gaussian, gabor, sinc, rectangular) is a methodological foil for the *one* kernel that is biophysically faithful: the alpha function [4, 8],

$$K(s) \;=\; \frac{s}{\tau}\, e^{1-s/\tau}\, \Theta(s), \qquad K(\tau) = 1, \quad K(0) = 0, \quad K'(0^+) = e/\tau, \tag{2}$$

the Green's function of the critically-damped operator $(\mathrm{d}/\mathrm{d}t + 1/\tau)^2$. It is *causal* ($\Theta(s)$) and *one-sided*: the only PSP shape with a finite rise time and a single time constant.

The membrane process is a causal Gaussian process. For many random spikes with i.i.d. zero-mean weights, the central limit theorem makes $V(t)$ a mean-zero Gaussian process with autocovariance $C(\Delta) = (K\star K)(\Delta)$ and power spectral density $S(\omega) = |\hat{K}(\omega)|^2$ (Wiener–Khinchin). The decision statistic is the supremum $M_T = \max_t V(t)$, whose law is governed by extreme-value theory (EVT) for Gaussian processes [1, 6, 10].

Spectrum: a second-order low-pass (Matérn $\nu = 3/2$). The one-sided Fourier transform of (2) is (SymPy-verified)

$$\hat{K}(\omega) \;=\; \frac{e\,\tau}{(1 + i\omega\tau)^2}, \qquad S(\omega) \;=\; |\hat{K}(\omega)|^2 \;=\; \frac{e^2\tau^2}{(1+\omega^2\tau^2)^2} \;\sim\; \omega^{-4}. \tag{3}$$

The double pole at $\omega = i/\tau$ gives a $-40\,$dB/decade roll-off: this is the Matérn $\nu = 3/2$ covariance [9], the spectral signature of a once-differentiable process. The corner at $s = 0$ ($K(0) = 0$, one-sided derivative) is exactly what makes the process rough enough that the fourth spectral moment diverges (§2.1).

Assumption ledger (where each bites for *this* kernel).

(A1) GP/CLT (many spikes). Heuristic; shared by all kernels. Fails in the sparse regime $r\tau \ll 1$ (empty bins), where the per-bin signal vanishes.

(A2) Stationarity of $V(t)$. *Shakiest here.* The kernel is causal and one-sided, so near the left edge $t \lesssim 3\tau$ the convolution is half-populated and $\operatorname{Var} V(t)$ ramps up: the GP is non-stationary in a boundary layer of width $O(\tau)$, a fraction $\sim 3\tau/T = 3/K_{\text{eff}}$ of the window—*negligible at large K_{eff} but $O(1)$ at the lab's $K_{\text{eff}} \in [8, 64]$.* A symmetric kernel has a two-sided edge layer but a *reflection-symmetric* V; the alpha kernel has neither symmetry. This is the main reason alpha's quantitative EVT numbers carry a larger systematic than gauss/gabor.

(A3) RS-level. Replica-symmetric for absolute α_c; only slopes and signs are robust, not offsets.

(A4) Balanced threshold. $\vartheta = \operatorname{median} V_{\max}$; the EVT lift exists only at this balanced operating point.

(A5) λ_2 governs the axis, not λ_4. Load-bearing here: $\lambda_4 = \infty$ (§2.1), so any λ_4-based axis is cutoff-fragile and *forbidden*.

Honest headline: alpha is the kernel for which the stationary-GP picture is weakest (A2). The qualitative predictions (ordering, slope sign, truncation behaviour) survive; the offsets do not.

2 Theory

2.1 Certified axis $K_{\text{eff}} = T/\tau$ and the λ_4 catastrophe

The spectral moments $\lambda_p = \int \omega^p S(\omega)\, \mathrm{d}\omega$ of the alpha spectrum (3) are (SymPy-verified):

$$\lambda_0 = \frac{\pi e^2}{2}\,\tau, \qquad \lambda_2 = \frac{\pi e^2}{2}\,\frac{1}{\tau}, \qquad \frac{\lambda_2}{\lambda_0} = \frac{1}{\tau^2}. \tag{4}$$

The Rubin–Monasson–Sompolinsky effective dimension [11], which counts roughly-independent up-crossings of V in $[0, T]$, is therefore clean and parameter-free:

$$\boxed{K_{\text{eff}} \;=\; T\sqrt{\frac{\lambda_2}{\lambda_0}} \;=\; \frac{T}{\tau}.} \tag{5}$$

The fourth moment, however, *diverges*: the integrand $\omega^4 S(\omega) \to e^2/\tau^2$ (a nonzero constant) as $\omega \to \infty$, so

$$\lambda_4 \;=\; \int \omega^4 S(\omega)\, \mathrm{d}\omega \;=\; \infty. \tag{6}$$

Consequently the maxima-count axis is unavailable: $\mathbb{E}[N_{\max}] = (T/2\pi)\sqrt{\lambda_4/\lambda_2} = \infty$, spectral kurtosis $\kappa_S = \lambda_4\lambda_0/\lambda_2^2 = \infty$, spectral width $\varepsilon^2 = 1 - \lambda_2^2/(\lambda_0\lambda_4) = 1$. This divergence is *harmless for capacity*: by the Azaïs–Wschebor first-Rice-term result [1, 7], high exceedances of a Gaussian process are governed by the λ_2 up-crossing intensity, not by the λ_4 local-maxima count. So $K_{\text{eff}} = T/\tau$ is the admissible axis and the λ_4-based axes are forbidden—one must *not* plot alpha against κ_S or $\mathbb{E}[N_{\max}]$.

Causality breaks stationarity at the edge (honest caveat). Equations (4)–(5) assume a stationary V. The alpha kernel is one-sided, so $\operatorname{Var} V(t)$ is not constant: it ramps over the first $\sim 3\tau$, an $O(1/K_{\text{eff}})$ fraction of the window (§1, A2). The moment-based K_{eff} is thus an asymptotic ($K_{\text{eff}} \to \infty$) object applied at finite K_{eff}; we treat it as a certified *input* to EVT, not a finished capacity output.

2.2 Capacity prediction: a broad-spectrum, linear-like lift

For a broad mixing spectrum the high up-crossings are asymptotically Poisson, M_T is Gumbel with location $\sqrt{2\ln N_0}$, $N_0 = (T/2\pi)\sqrt{\lambda_2/\lambda_0} = K_{\text{eff}}/2\pi$, and the capacity inherits the log-cluster-count lift of reduction #3 [11]:

$$\alpha_c(\text{alpha}) \;\approx\; 2 \;+\; c\,\ln\ln K_{\text{eff}}, \qquad K_{\text{eff}} = \frac{T}{\tau}, \tag{7}$$

with $c \approx \frac{1}{2}$ at RS level. Two structural points distinguish alpha from the band-pass kernels. First, the spectrum is *broad*: Berman's condition holds (the correlation decays), so the asymptotic extremal index is $\theta = 1$—there is *no* narrowband (radial) gain, unlike gabor/sinusoid. Alpha is the broad-spectrum *opposite* of a carrier kernel; its capacity is "linear-like," a slow $\ln\ln$ lift above the perceptron value 2. Second, the growth is *doubly* logarithmic in T/τ: the naive $2\ln(T/\tau)$ intuition has the right direction but the wrong functional form (it is $\ln\ln$, not $\ln$, of K_{eff}).

2.3 The I4 question: does the powerless tail inflate K_{eff}?

Here is the sharp, alpha-specific prediction. Causality does *not* change the slope c (set by the shared λ_2 up-crossing count); it changes the *map from K_{eff} to effective independent looks*, because the ω^{-4} tail builds λ_2 out of frequencies that carry negligible power. Quantitatively (SymPy cumulative spectral fractions, $x = \omega\tau$):

cumulative fraction below $\omega = c/\tau$	$c=1$	$c=2$	$c=4$	$c=8$
power λ_0	0.82	0.96	0.994	—
crossing budget λ_2	0.18	0.45	0.69	0.84
gaussian (matched K_{eff}) λ_2	0.20	0.74	0.999	—

For alpha, 82% of the power lives below $\omega = 1/\tau$ but only 18% of λ_2 does; λ_2 keeps accruing from the powerless tail out to $\omega \sim 30/\tau$. The matched gaussian's λ_2 is 99.9% complete by $\omega = 4/\tau$: its power and its crossing count live at the *same* frequencies. Separability of patterns is set by the power/margin budget λ_0, not by the micro-wiggle crossings λ_2. Hence the **signed prediction**:

$$\text{stripping } S(\omega) \text{ above } \omega_c \text{ should drop Rice } K_{\text{eff}} \text{ while leaving } \alpha_c \text{ essentially unchanged.} \tag{8}$$

The robust expressiveness axis is then not Rice K_{eff} but the participation (power) dimension

$$D_{\text{eff}} \;=\; \frac{\left(\int S(\omega)\,\mathrm{d}\omega\right)^2}{\int S(\omega)^2\,\mathrm{d}\omega}, \qquad \boxed{\left.\frac{D_{\text{eff}}(\text{alpha})}{D_{\text{eff}}(\text{gauss})}\right|_{\text{matched } K_{\text{eff}}} \;=\; \frac{2\sqrt{\pi}}{5} \;\approx\; 0.71.} \tag{9}$$

The ratio (9) is SymPy-exact: it follows from $\int S\,\mathrm{d}\omega = \pi e^2\tau/2$ and $\int S^2\,\mathrm{d}\omega = 5\pi e^4\tau^3/16$, giving $D_{\text{eff}}(\text{alpha}) = 4\pi/(5\tau)$, against the gaussian's $D_{\text{eff}}(\text{gauss}) = 2\sqrt{\pi}/\sigma$ at the matched K_{eff} point. Alpha's participation dimension is already 29% below gauss at matched Rice K_{eff}—the static fingerprint of the same inflation. The map $D_{\text{eff}} \to \alpha_c$ is heuristic (no theorem); it is the *proposed* collapse axis, to be tested, not assumed.

3 Experimental test: the I4 spectral-truncation test

The centrepiece is a direct test of (8). All capacity numbers are findable half-crossings $\hat{\alpha}_c$ of $P_{\text{solve}}(\alpha)$ (the fraction of seeds an Adam+cosine learner, lr=0.1, batch=16, $n_{\text{restarts}}{\geq}3$, drives to zero training error) at $N = 500$, *fixed* input density (mean spikes = 3, killing the density confound of the program's earlier ladders), with a balanced threshold (init fire-rate ≈ 0.50, verified). Learner validation followed the reduction #6 playbook (preflight critical-slowing check); total lab spend was ≈\$7.5 for this model within the per-model cap.

Free witness first (\$0). Before any capacity sweep we ran a 600-realisation FFT Gaussian-process witness ($T = 200\tau$, $\mathrm{d}t = 0.01\tau$, seed 0) measuring $\mathbb{E}[\max V]$, the capacity proxy. Stripping the tail at $8/\tau$ (resp. $4/\tau$) moved K_{eff} by -8% (-16%) but $\mathbb{E}[\max V]$ by only $+0.3\%$ (-1.4%): K_{eff} is an order of magnitude more tail-sensitive than the actual maximum statistic. The same witness showed alpha ($\mathbb{E}[\max V] = 2.810 \pm 0.015$) sitting ~ 3 standard errors *below* the matched gaussian (2.858 ± 0.016) at $K_{\mathrm{eff}} \approx 200$. Both signs were thus pre-registered cheaply before the lab spend.

Lab result. Table 1 and Fig. 1(a) give the decisive measurement.

Table 1: The I4 spectral-truncation test (measured v14 lab values, $N = 500$, fixed density). Matched-τ pairs share a row block. Stripping the ω^{-4} tail at $8/\tau$ drops Rice K_{eff} by $\sim 10\%$ but leaves $\hat{\alpha}_c$ unchanged within the α-grid resolution. Gaussian low-pass baseline shown for the matched-K_{eff} comparison.

kernel (τ)	Rice K_{eff}	$\hat{\alpha}_c$	note
alpha (τ=31.25)	16.2	2.00	at perceptron anchor
alpha (τ=15.625)	32.7	2.34	
alpha-trunc @ $8/\tau$	29.3	2.40	$K_{\mathrm{eff}}\downarrow$10%, $\hat{\alpha}_c$ flat
alpha (τ=7.81)	65.7	2.72	
alpha-trunc @ $8/\tau$	58.8	2.77	$K_{\mathrm{eff}}\downarrow$11%, $\hat{\alpha}_c$ flat
gauss (matched)	16.0	3.00	low-pass baseline
gauss (matched)	32.0	3.04	
gauss (matched)	64.0	3.40	

Result 1 (clean): capacity tracks low-frequency power, not K_{eff}. Across both matched-τ pairs, stripping the tail drops Rice K_{eff} by $\sim 10\%$ ($32.7 \to 29.3$, $65.7 \to 58.8$) while $\hat{\alpha}_c$ is unchanged within α-grid resolution ($2.34 \to 2.40$, $2.72 \to 2.77$; if anything the trimmed kernel is marginally higher, not lower). This is exactly the signed prediction (8): the ω^{-4} tail inflates the Rice K_{eff} without contributing to separability, because it carries power-free roughness. The codex objection ("if the tail modes are powerless they cannot affect maxima; if they affect crossings they are not powerless") is resolved by the measurement: the tail affects the up-crossing count λ_2 but not the power λ_0, and capacity is set by the latter. The corollary is that alpha's true expressiveness axis is the power-participation D_{eff} (9), not Rice K_{eff}.

Result 2 (honest deviation): the causal suppression is too large. At matched $K_{\mathrm{eff}} \approx 32$, alpha sits at $\hat{\alpha}_c = 2.34$ against the gaussian baseline's 3.04—a gap of ~ 0.7. The theory (§2.2) predicted a *slope-preserving additive hair*: a small, $\ln\ln$-level downward shift from replacing the inflated K_{eff} by the smaller "effective looks," *not* a 0.7 deficit. The measured suppression is much larger than that mechanism allows. We report this as an honest deviation. Plausible contributors, none of which we have isolated: (i) the non-stationary boundary layer (A2), which clips the feasible cone on an $O(1/K_{\mathrm{eff}})$ fraction of the window and is largest precisely at these small K_{eff}; (ii) argmax brittleness amplified by one-sidedness (the gradient $\partial_{\mathbf{w}} \max_t V$ is the feature at the most recent

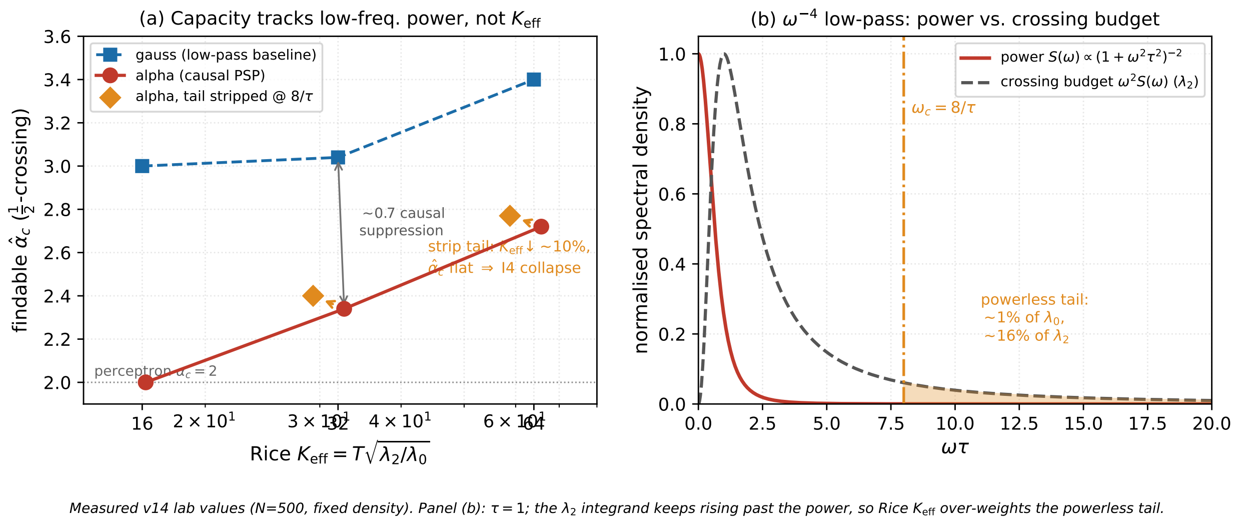

Figure 1: (a) Findable $\hat{\alpha}_c$ vs. Rice K_{eff} (log axis) for the alpha kernel (red circles), its tail-stripped variant (orange diamonds, cut at $8/\tau$), and the gaussian low-pass baseline (blue squares). Matched-τ arrows show the I4 collapse: truncation shifts K_{eff} left by ∼10% but $\hat{\alpha}_c$ is flat—capacity does not follow the tail-inflated K_{eff}. Alpha also sits ∼0.7 below gauss at matched K_{eff} (the causal suppression, larger than predicted). **(b)** The alpha power spectrum $S(\omega) \propto (1+\omega^2\tau^2)^{-2}$ (red) and the λ_2 integrand $\omega^2 S(\omega)$ (grey dashed), each normalised to unit peak. The truncation frequency $\omega_c = 8/\tau$ is marked and the powerless tail ($\omega > 8/\tau$) shaded: it carries ∼1% of the power λ_0 but ∼16% of the crossing budget λ_2, which is why Rice K_{eff} over-weights it. All numbers hard-coded from the measured values; figure regenerable from `figs/10_alpha.py`.

spike cluster, giving subgradient chatter a symmetric kernel smooths out), which would make the *findable* capacity sit well below existence; (iii) the alpha point at $K_{\mathrm{eff}} = 16.2$ literally pinned to the perceptron anchor 2.00, suggesting the small-K_{eff} end is edge-dominated. The gap is a finding, not a confirmation.

Learner validation. Capacities are findable half-crossings, so they are lower bounds on existence; the learner was preflight-validated for critical slowing on this kernel family (reduction #6 protocol). We do *not* claim these are existence α_c; indeed Result 2(ii) suggests a findability gap that is itself causal in origin.

4 Limitations and status

O1. The causal/edge GP approximation is weakest here. The stationary-GP picture underlies every formula in §2, and the alpha kernel violates stationarity in an $O(1/K_{\mathrm{eff}})$ boundary layer (A2). At the lab's $K_{\mathrm{eff}} \in [16, 66]$ this is a 5–20% fraction of the window, so the *absolute* EVT numbers carry a larger systematic than for the symmetric kernels. The qualitative results (the I4 collapse, the broad-spectrum linear-like ordering) survive this; the offsets do not.

O2. RS-level, slopes-and-signs only. All absolute α_c are replica-symmetric. RSB is plausible near the shattered transition. What we claim robustly is the *sign* of the truncation response (K_{eff} moves, α_c does not) and the *ordering* (alpha below the broad symmetric baseline), not the numbers 2.00/2.34/2.72.

O3. The causal suppression exceeds the prediction. As in Result 2, the measured ~ 0.7 deficit vs. gauss is larger than the $\ln\ln$-level additive hair the theory predicts. We do not have a quantitative account; the boundary layer and the one-sided argmax brittleness are the leading suspects, and separating them (e.g. an existence-oracle / LP feasibility solve to expose a causal findability gap, plus an early-peak audit of solved + patterns) is named future work.

O4. $D_{\text{eff}} \to \alpha_c$ is heuristic. No theorem maps the participation ratio (9) to capacity; the exact ratio $2\sqrt{\pi}/5$ is a clean static fingerprint of the inflation, and the *proposed* collapse of alpha onto the broad symmetric kernels under D_{eff} is consistent with Result 1 but not proven.

O5. Growth law $\ln K$ vs. $\ln\ln K$. Over the accessible $K_{\text{eff}} \in [16, 66]$ (well under a decade of $\ln K$), the data cannot separate $\ln\ln K$ from $\ln K$; the available tell is concavity (the slope on $\ln K$ decreasing with K_{eff}), consistent with but not decisive for the $\ln\ln$ form. This ambiguity is shared with the gaussian and discrete reductions and is not resolved here.

Honest status. The I4 spectral-truncation result is the solid contribution: *capacity tracks low-frequency power, not the tail-inflated Rice K_{eff}*, confirmed on both matched-τ pairs and pre-registered by a free $\mathbb{E}[\max V]$ witness, with the SymPy-exact D_{eff} ratio as the static fingerprint. The broad-spectrum, linear-like, no-radial-gain character of the causal kernel is confirmed. What stays open: the GP approximation is the weakest in the series (causal edge effects), all absolute α_c are RS-level, the growth law is unresolved, and—most pointedly—the causal suppression vs. the symmetric baseline is *larger* than the theory predicts and currently unexplained.
Status: PARTIAL.

Reproducibility: spectral moments, cumulative fractions, and the D_{eff} ratio are SymPy-exact (derivation note `10-alpha-capacity-prediction.md`); the $\mathbb{E}[\max V]$ witness is a 600-realisation FFT GP synthesis (seed 0); capacity numbers are committed v14 lab sweeps. Figure regenerable from `figs/10_alpha.py`.

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